A Four-Term Counterexample to the Special Image Conjecture in Three Pairs

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Abstract

We give an explicit counterexample to the Special Image Conjecture in three contraction pairs. In $\mathbb{C}[\tau, w, v, t, z, y]$, let

$$f = \tau(t - y)(wz + vt), \qquad g = y.$$

For the standard contraction map $\mathcal{E}$, we prove for every $m \geq 1$ that

$$\mathcal{E}(f^m) = 0, \qquad [t]\mathcal{E}(gf^m) = (-1)^{m-1}(m + 1)!\, m!.$$

Thus every positive power of f lies in the image of the operators $\partial_t - \tau, \partial_z - w, \partial_y - v$, whereas $gf^m \notin \mathcal{M}_3$ for every $m \geq 1$. The polynomial f has four terms and the multiplier is linear. Its two-pair dehomogenization is exactly Christopher D. Long's counterexample to the Mathieu conjecture for $SU(2)$; the present construction is its one-pair bihomogeneous lift. Consequently the minimum number of pairs in which the Special Image Conjecture can fail is either two or three.

1 The counterexample

For $r \geq 1$, write

$$R_r = \mathbb{C}[\zeta_1, \ldots, \zeta_r, z_1, \ldots, z_r], \qquad \mathcal{M}_r = \sum_{i=1}^{r} (\partial_{z_i} - \zeta_i)R_r.$$

The Special Image Conjecture asserts that $\mathcal{M}_r$ is a Mathieu subspace: if $h^m \in \mathcal{M}_r$ for every $m \geq 1$, then $qh^m \in \mathcal{M}_r$ for every $q \in R_r$ and all sufficiently large m; see Zhao [1] and van den Essen, Wright, and Zhao [2].

Define the polynomial-valued contraction map

$$\mathcal{E}_r : R_r \longrightarrow \mathbb{C}[z_1, \ldots, z_r], \qquad \mathcal{E}_r(\zeta^\alpha z^\beta) = \partial_z^\alpha z^\beta.$$

Lemma 1. *For every $r \geq 1$, one has $\mathcal{M}_r = \ker \mathcal{E}_r$.*

Proof. For $h \in R_r$,

$$\mathcal{E}_r(\partial_{z_i} h) = \partial_{z_i} \mathcal{E}_r(h) = \mathcal{E}_r(\zeta_i h).$$

Thus $\mathcal{E}_r((\partial_{z_i} - \zeta_i)h) = 0$, so $\mathcal{M}_r \subseteq \ker \mathcal{E}_r$. Conversely,

$$\zeta_i h \equiv \partial_{z_i} h \pmod{\mathcal{M}_r}.$$

Repeatedly applying this congruence gives, for every monomial,

$$\zeta^\alpha z^\beta \equiv \partial_z^\alpha z^\beta = \mathcal{E}_r(\zeta^\alpha z^\beta) \pmod{\mathcal{M}_r}.$$

By linearity, $p \equiv \mathcal{E}_r(p) \pmod{\mathcal{M}_r}$. Since $\mathcal{E}_r$ is the identity on $\mathbb{C}[z_1, \ldots, z_r]$, $\mathcal{E}_r(p) = 0$ implies $p \in \mathcal{M}_r$. □

Use the pairs

$$(\tau, t), \qquad (w, z), \qquad (v, y)$$

and put

$$\mathcal{M}_3 = (\partial_t - \tau)R_3 + (\partial_z - w)R_3 + (\partial_y - v)R_3.$$

Theorem 2. *Let*

$$\boxed{f = \tau(t - y)(wz + vt), \qquad g = y.}$$

Then, for every $m \geq 1$,

$$\mathcal{E}_3(f^m) = 0, \qquad [t]\mathcal{E}_3(gf^m) = (-1)^{m-1}(m+1)!\, m! \neq 0.$$

In particular, $\mathcal{E}_3(gf^m) \neq 0$. Consequently $f^m \in \mathcal{M}_3$ and $gf^m \notin \mathcal{M}_3$ for every $m \geq 1$.

In expanded form,

$$\boxed{f = \tau vt^2 - \tau vty + \tau wtz - \tau wyz.}$$

This displays the advertised four terms directly.

2 Proof

First omit the pair (τ, t). On $\mathbb{C}[W, Z, V, Y]$, define

$$\mathcal{F}(W^a Z^b V^c Y^d) = \delta_{a,b}\delta_{c,d}\, a!\, c!$$

and set

$$P = (1 - Y)(WZ + V), \qquad Q = Y.$$

Expanding the second factor gives

$$P^m = \sum_{k=0}^{m} \binom{m}{k} W^{m-k} Z^{m-k} V^k (1 - Y)^m.$$

For a nonzero contraction, the V, Y pair selects $[Y^k](1 - Y)^m = (-1)^k \binom{m}{k}$. Hence

$$\mathcal{F}(P^m) = \sum_{k=0}^{m} (-1)^k \binom{m}{k}^2 (m - k)!\, k! = m! \sum_{k=0}^{m} (-1)^k \binom{m}{k} = 0. \tag{1}$$

After multiplication by $Q = Y$, the V, Y pair instead selects $[Y^k]Y(1 - Y)^m = (-1)^{k-1}\binom{m}{k-1}$ for $1 \leq k \leq m$. Therefore

$$\begin{aligned} \mathcal{F}(YP^m) &= m! \sum_{k=1}^{m} (-1)^{k-1} \binom{m}{k-1} \\ &= (-1)^{m-1} m!. \end{aligned} \tag{2}$$

Homogenize in the coordinate variables Z, Y, treating W, V as coefficient, or dual, variables:

$$\widetilde{P}(t,w,z,v,y) = t^2 P\left(w, \frac{z}{t}, v, \frac{y}{t}\right) = (t-y)(wz+vt).$$

Each factor of P has total W,V-degree one. Every monomial of P^m that survives $\mathcal{F}$ therefore has total (Z,Y)-degree m, matching its total W,V-degree. Its coordinate homogenization contains $t^{2m-m} = t^m$, exactly matching the power τ^m. The new contraction contributes $m!$, and (1) gives

$$\mathcal{E}_3(f^m) = m!\mathcal{F}(P^m) = 0.$$

For YP^m, a surviving term of P^m has total (Z,Y)-degree $m-1$. Its homogenization contains $t^{2m-(m-1)} = t^{m+1}$. Contracting against τ^m leaves one power of t and contributes $(m+1)!$. By (2),

$$[t]\mathcal{E}_3(yf^m) = (m+1)!\mathcal{F}(YP^m) = (-1)^{m-1}(m+1)!\,m! \neq 0.$$

Lemma 1 completes the proof of Theorem 2.

3 Relation with the xz- and $SU(2)$ counterexamples

Write the coordinate functions on $SU(2)$ as

$$\begin{pmatrix} a & c \\ b & d \end{pmatrix}.$$

Under the substitution

$$W = a, \qquad Z = d, \qquad V = b, \qquad Y = -c,$$

the two-pair seed and multiplier become

$$P = (1-Y)(WZ+V) \longmapsto (1+c)(ad+b), \qquad Q = Y \longmapsto -c.$$

This is precisely Long's $SU(2)$ counterexample [3]. Indeed normalized Haar integration satisfies

$$\int_{SU(2)} a^r b^s c^t d^u \, dg = (-1)^s \delta_{r,u}\delta_{s,t} \frac{r!\,s!}{(r+s+1)!}.$$

Because every monomial of P^m has total W,V-degree m, this formula gives

$$\mathcal{F}(P^m) = (m+1)! \int_{SU(2)} ((1+c)(ad+b))^m \, dg$$

and the analogous identity with multipliers Y and $-c$. Thus equations (1)–(2) recover Long's moments

$$\int_{SU(2)} ((1+c)(ad+b))^m \, dg = 0, \qquad \int_{SU(2)} (-c)((1+c)(ad+b))^m \, dg = \frac{(-1)^{m-1}}{m+1}.$$

Under the Müger–Tuset integration substitution used by Long,

$$(a,b,c,d) = \big((1-x)z_2, xz_1, -z_1^{-1}, z_2^{-1}\big),$$

the same seed becomes

$$(1-z_1^{-1})\big((1-x)+xz_1\big),$$

which is Long's three-term counterexample to $xz(1,1)$. Accordingly, the seed and its xz- and $SU(2)$-interpretations are due to Long. The contribution here is the degree-two one-pair bihomogenization that produces the four-term counterexample to the Special Image Conjecture in three pairs.

Corollary 3. *If r_{SIC} denotes the minimum number of contraction pairs in which the Special Image Conjecture fails, then*

$$2 \leq r_{\mathrm{SIC}} \leq 3.$$

Proof. The upper bound is Theorem 2. The one-pair case follows from [2, Theorem 2.8]: take the $\mathbb{Q}$-algebra $A = \mathbb{C}[\tau]$ and $a = \tau$. Then a is a non-zero-divisor and the principal ideal $Aa = (\tau)$ is radical, precisely the hypotheses of that theorem. $\square$

4 Size, scope, and verification

The coefficients of f lie in $\{1, -1\}$. With respect to (τ, w, v)-degree and (t, z, y)-degree, f has bidegree $(2, 2)$ and ordinary degree four; $g = y$ has bidegree $(0, 1)$. The polynomial f is not of the separated form $\lambda(\zeta)P(z)$; consequently the example does not give a three-variable Generalized Vanishing Conjecture counterexample.

Thus the known one-pair result and Theorem 2 give

$$2 \leq r_{\mathrm{SIC}} \leq 3.$$

We make no claim here concerning the two-pair case or absolute literature priority.

The all-order proof above is independent of computation. An exact sparse-arithmetic checker expands f, verifies the contractions through $m = 10$, and checks the two binomial identities through $m = 99$. The checker is included in the accompanying Zenodo record and in the companion repository.[1] As a low-order checksum,

$$\mathcal{E}_3(f) = 0, \qquad \mathcal{E}_3(yf) = 2t - y.$$

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References

[1] W. Zhao, *Images of commuting differential operators of order one with constant leading coefficients*, J. Algebra **324** (2010), 231–247. doi:10.1016/j.jalgebra.2010.04.022.

[2] A. van den Essen, D. Wright, and W. Zhao, *On the Image Conjecture*, J. Algebra **340** (2011), 211–224. doi:10.1016/j.jalgebra.2011.04.036.

[3] C. D. Long, *Counterexamples to the xz-Conjecture and the Mathieu Conjecture for SU(2)*, arXiv:2607.19012v1 (2026). arXiv:2607.19012.

[1] Repository path: `scripts/verify_three_pair_image_mathieu_counterexample.py`.